

ABHAY MADAAN

Generative AI Engineer — Python & Full Stack Developer

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PROFESSIONAL SUMMARY

Generative AI Engineer with 2+ years building production AI systems in Python — RAG pipelines, multi-agent orchestration, and multimodal search. Delivered 10+ applications across iGaming, AI agents, and fashion-tech using LangChain, LangGraph, HuggingFace, and FastAPI. Improved backend performance by up to 70% through optimization and scalable architecture.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C/C++

Generative AI & ML: RAG Pipelines, LangChain, LangGraph, Multi-Agent Systems, HuggingFace (IDM-VTON, CLIP, sentence-transformers), Weaviate, Groq (Llama 3.3 70B), Multimodal Search

Backend & APIs: FastAPI, RESTful APIs, Celery, Asynchronous Python, Node.js, Express.js, Socket.IO, BullMQ

Frontend: React.js, Next.js

Databases & Caching: PostgreSQL, MongoDB, Redis

DevOps & Tools: Docker, AWS (EC2, S3, ECS, ALB, ASG), Scrapy, Selenium, Git, Linux

KEY PROJECTS

UrFitAI – AI Fashion Discovery Platform — FastAPI, Python, HuggingFace, Next.js, RAG, Weaviate

- Engineered a **RAG pipeline** using **LangChain** and **Groq (Llama 3.3 70B)** — retrieving semantically matched products from **Weaviate** via **CLIP** and **sentence-transformer** embeddings for grounded, product-aware styling recommendations
- Built an **AI virtual try-on** feature integrating **IDM-VTON** (HuggingFace Spaces) with async background job architecture, Cloudinary storage, and real-time polling
- Built a configurable **Scrapy/Selenium** scraping pipeline with stealth patching and JSONB-driven selectors — cataloging **100,000+ products** across multiple brands for NLP-based product matching

AI Agent Platform – Multi-Agent Research Assistant — Python, LangGraph, OpenAI, Anthropic, Docker

- Implemented a **multi-agent research assistant** using **LangGraph** with supervisor-routed orchestration across 3 specialized agents (Researcher, Writer, Reviewer) and **tool calling** for automated web search
- Configured **multi-provider LLM support** (OpenAI, Anthropic) with runtime switching, and an iterative writer-reviewer refinement loop (up to 3 revision cycles) with human-in-the-loop review checkpoints
- Containerized the application with **Docker** for reproducible deployment

Betski – iGaming Platform — Node.js, PostgreSQL, Redis, BullMQ

- Collaborated within a **10-engineer** team to develop core platform features (**VIP Levels**, **Campaign Management**, **Bonus Systems**, **Race Engine**), reducing public API latency by up to **80%** through query tuning and **Redis** caching on a platform serving **10,000+** average daily active users.

PROFESSIONAL EXPERIENCE

Solution Engineer — Gammastack, Bangalore

Nov 2025 – Present

- Develop and maintain high-scale iGaming platforms using **Node.js**, **BullMQ**, **Redis**, and **PostgreSQL**, optimizing public APIs via query tuning and caching — reducing **p95 response times by 35%**
- Delivered business-critical features including **VIP Levels**, **Campaign Management**, **Bonus Systems**, and **Race Engine** within a **10-engineer** team, driving engagement on a platform serving **10,000+** average daily active users
- Concurrently building AI-powered applications using **LangChain**, **LangGraph**, **RAG pipelines**, **Weaviate**, **Groq (Llama 3.3 70B)**, and **CLIP** for multimodal search and product recommendations

Software Developer — 75 Way Technologies Pvt. Ltd., Mohali

Jan 2024 – May 2025

- Developed scalable full-stack applications using **Node.js**, **React.js**, **React Native**, and **PostgreSQL** across multiple business domains
- Reduced backend response time by up to **70%** through query optimization, caching, and asynchronous workflow implementation
- Integrated **Stripe** payment gateway with secure webhook validation and complete payment lifecycle handling

ACHIEVEMENTS & CERTIFICATIONS

Certifications: Design and Analysis of Algorithms, Ultimate AWS Certified Developer Associate 2025 (DVA-C02) —

Achievements: Solved 700+ problems (Level 7, Coding Ninjas), Ranked 11th in Code Kaze

EDUCATION

Bachelor of Technology in Computer Science

Guru Nanak Dev Engineering College, Ludhiana

2020 – 2024

CGPA: 8.3/10